

Factorial

FACTORIAL

Write a Java program to calculate the factorial of a given number using loops.

```
1 import java.util.Scanner;
2 class factorial{
3     public static void main(String[] args){
4         Scanner sc= new Scanner(System.in);
5         int n=sc.nextInt();
6         int fact=1;
7         for(int i=1;i<=n;i++){
8             fact=fact*i;
9         }
10        System.out.println(fact);
11    }
12 }
```

OUTPUT

24

QUESTIONS 25 / 31 completed

Prime **PRIME**

Write a Java program to check whether a given number is a prime number or not using loops and conditional statements.

PROGRAM EDITOR

Java 

Run

Save

```
1 import java.util.Scanner;
2 class prime_number{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int n=sc.nextInt();
6         int p=0;
7         for(int i=1;i<=n;i++){
8             if(n%i==0){
9                 p=p+1;
10            }
11        }
12    }
13    if(p==2){
14        System.out.println("Prime number");
15    }
16    else{
17        System.out.println("Not a prime");
18    }
19 }
20
21 }
```

OUTPUT

Prime number

INPUT

11



Main.java

```
1+ class composite{
2+   public static void main(String[] args){
3+       int n=4;
4+       int t=0;
5+       for(int i=1;i<=n;i++){
6+           if(n%i==0){
7+               t=t+1;
8+           }
9+       }
10+       if(t>2){
11+           System.out.println("COMPOSITE NUMBER");
12+       }
13+       else{
14+           System.out.println("NOT A COMPOSITE NUMBER");
15+       }
16+   }
17+ }
18+ }
19+ }
20
```

Output

COMPOSITE NUMBER

=== Code Execution Successful ===

Clear



QUESTIONS 25 / 31 completed

Leap **LEAP**

Write a Java program to check whether a given year is a leap year or not using conditional statements.

PROGRAM EDITOR

Java 

Run

Save

```
1 import java.util.Scanner;
2 class leap_year{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int year=sc.nextInt();
6         if((year%4==0 && year%100!=0) || (year%400==0)){
7             System.out.println("Leap Year");
8         }
9         else{
10            System.out.println("Not a Leap Year");
11        }
12    }
13 }
14
```

OUTPUT

Leap Year

INPUT

2008

QUESTIONS 25 / 31 completed

Armstrong

ARMSTRONG

Write a Java program to check whether a given number is an Armstrong number or not using loops and conditional statements.

PROGRAM EDITOR

Java

Run

Save

```
1 import java.util.Scanner;
2 class armstrong{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         System.out.println("Enter the number:");
6         int n=sc.nextInt();
7         int temp=n;
8         int sum=0;
9         while(n>0){
10             int digit=n%10;
11             sum=sum+(digit*digit*digit);
12             n=n/10;
13         }
14         if(sum==temp){
15             System.out.println("Armstrong number");
16         }
17         else{
18             System.out.println("Not a Armstrong number");
19         }
20     }
21 }
```

OUTPUT

```
Enter the number:
Armstrong number
```

INPUT

153

Fibonacci **FIBONACCI**

Write a Java program to print the Fibonacci series up to a given number using loops.

```
1 import java.util.Scanner;
2 class fibonacci{
3     public static void main(String[] args){
4         Scanner sc=new Scanner(System.in);
5         int n=sc.nextInt();
6         int a=0;
7         int b=1;
8         for(int i=1;i<=n;i++){
9             System.out.print(a + " ");
10            int next=a+b;
11            a=b;
12            b=next;
13        }
14    }
15 }
16 }
```

OUTPUT

```
0 1 1 2 3
```

5

Main.java

```
1+ class composite{
2+     public static void main(String[] args){
3+         int n=1234;
4+         int rev=0;
5+         while(n>0){
6+             int digit=n%10;
7+             rev=(rev*10)+digit;
8+             n=n/10;
9+         }
10      System.out.println(rev);
11  }
12 }
```

Output

4321
=== Code Execution Successful ===

Optum

Now hiring for *accounts*
receivable roles

Apply today

Main.java

```
1+ class factors_of_a_number{
2+     public static void main(String[] args){
3+         int n=26;
4+         System.out.println("Factors of a number:");
5+         for(int i=1;i<=n;i++){
6+             if(n%i==0){
7+                 System.out.println(i);
8+             }
9+         }
10+     }
11+ }
```

Output

Factors of a number:
1
2
13
26

=== Code Execution Successful ===



Main.java

```
1+ class factors_of_a_number{
2+     public static void main(String[] args){
3+         int n=6;
4+         int sum=0;
5+         System.out.println("sum of n numbers:");
6+         for(int i=1;i<=n;i++){
7+             sum=sum+i;
8+         }
9+         System.out.println(sum);
10        }
11    }
12 }
```

Output

sum of n numbers:
21
=== Code Execution Successful ===

RT-Pattern **RT-PATTERN**

Write a Java program to
print the following
pattern using loops.*

```
**  
***  
****  
*****
```

```
1 class main{  
2     public static void main(String[] args){  
3         int n=5;  
4         for(int i=1;i<=n;i++){  
5             for(int j=1;j<=i;j++){  
6                 System.out.print("*");  
7             }  
8             System.out.println();  
9         }  
10    }  
11 }
```

OUTPUT

```
*  
**  
***  
****  
*****
```

Your INPUT goes
here!

LT-Pattern **LT-PATTERN**

Write a Java program to
print the following
pattern using loops:*****

**

*

```
1 class pattern{
2     public static void main(String[] args){
3         int n=5;
4         for(int i=n;i>=1;i--){
5             for(int j=1;j<=i;j++){
6                 System.out.print("*");
7             }
8
9             System.out.println();
10        }
11    }
12 }
13
```

OUTPUT

```
*****
****
***
**
*
```

Your INPUT goes
here!



Main.java

```
1 class Main {  
2     public static void main(String[] args) {  
3         int[] arr = {1, 2, 3, 2, 4, 5, 3, 6};  
4  
5         System.out.print("Array after removing duplicates: ");  
6  
7         for (int i = 0; i < arr.length; i++) {  
8             boolean duplicate = false;  
9  
10  
11             for (int j = 0; j < i; j++) {  
12                 if (arr[i] == arr[j]) {  
13                     duplicate = true;  
14                     break;  
15                 }  
16             }  
17  
18  
19             if (!duplicate) {  
20                 System.out.print(arr[i] + " ");  
21             }  
22         }  
23     }  
24 }
```

Output

Array after removing duplicates: 1 2 3 4 5 6
=== Code Execution Successful ===

Clear

Main.java

```
1+ class Main {  
2+     public static void main(String[] args) {  
3+         int n = 7;  
4+  
5+         int a = 0, b = 1, c = 0;  
6+  
7+         if (n == 1) {  
8+             System.out.println(a);  
9+         } else if (n == 2) {  
10+             System.out.println(b);  
11+         } else {  
12+             for (int i = 3; i <= n; i++) {  
13+                 c = a + b;  
14+                 a = b;  
15+                 b = c;  
16+             }  
17+             System.out.println("Nth Fibonacci term = " + c);  
18+         }  
19+     }  
20+ }
```

Run

Output

Clear

Nth Fibonacci term = 8

=== Code Execution Successful ===

Optum

Now hiring for *accounts*
receivable roles

Apply today

Main.java

```
1 class Main {
2     public static void main(String[] args) {
3         int[] arr = {1, 2, 2, 3, 4, 3, 2, 1};
4
5         for (int i = 0; i < arr.length; i++) {
6
7             boolean counted = false;
8             for (int j = 0; j < i; j++) {
9                 if (arr[i] == arr[j]) {
10                     counted = true;
11                     break;
12                 }
13             }
14             if (counted) {
15                 continue;
16             }
17
18             int count = 1;
19             for (int j = i + 1; j < arr.length; j++) {
20                 if (arr[i] == arr[j]) {
21                     count++;
22                 }
23             }
24             System.out.println(arr[i] + " -> " + count);
25         }
26     }
27 }
```

Output

```
1 -> 2
2 -> 3
3 -> 2
4 -> 1
=== Code Execution Successful ===
```

Main.java

```
1+ class Main {  
2+     public static void main(String[] args) {  
3+         int n = 5;  
4+  
5+         for (int i = 1; i <= n; i++) {  
6+             for (int j = 1; j <= n - i; j++) {  
7+                 System.out.print(" ");  
8+             }  
9+  
10+            for (int j = 1; j <= (2 * i - 1); j++) {  
11+                System.out.print("*");  
12+            }  
13+  
14+            System.out.println();  
15+        }  
16+    }  
17+ }
```

Output

```
*  
***  
*****  
*****  
*****  
*****  
=== Code Execution Successful ===
```

Main.java

```
1+ class Main {  
2+     public static void main(String[] args) {  
3         int n = 5;  
4         for (int i = 1; i <= n; i++) {  
5             for (int j = 1; j <= i; j++) {  
6                 System.out.print("**");  
7             }  
8             System.out.println();  
9         }  
10        for (int i = n - 1; i >= 1; i--) {  
11            for (int j = 1; j <= i; j++) {  
12                System.out.print("**");  
13            }  
14            System.out.println();  
15        }  
16    }  
17 }
```

Output

```
*  
**  
***  
****  
*****  
*****  
****  
***  
**  
*  
  
=== Code Execution Successful ===
```

Main.java

```
1+ class Main {  
2+     public static void main(String[] args) {  
3+         int n = 5;  
4+         for (int i = n; i >= 1; i--) {  
5+             for (int j = 1; j <= n - i; j++) {  
6+                 System.out.print(" ");  
7+             }  
8+             for (int j = 1; j <= 2 * i - 1; j++) {  
9+                 System.out.print("*");  
10+             }  
11+             System.out.println();  
12+         }  
13+     }  
14+ }
```

Output

```
*****  
*****  
*****  
***  
*  
  
=== Code Execution Successful ===
```

Main.java

```
1 class Main {  
2     public static void main(String[] args) {  
3         int n = 5;  
4  
5         for (int i = 1; i <= n; i++) {  
6             for (int j = 1; j <= n; j++) {  
7  
8                 if (i == 1 || i == n || j == 1 || j == n) {  
9                     System.out.print("* ");  
10                } else {  
11                    System.out.print(" ");  
12                }  
13            }  
14            System.out.println();  
15        }  
16    }  
17 }
```

Output

```
* * * * *  
*       *  
*       *  
*       *  
* * * * *
```

=== Code Execution Successful ===

Clear

Main.java

```
1+ class Main {
2+     public static void main(String[] args) {
3+         int[][] a = {
4+             {1, 2},
5+             {3, 4}
6+         };
7+         int[][] b = {
8+             {5, 6},
9+             {7, 8}
10+        };
11+        int[][] c = new int[2][2];
12+
13+        for (int i = 0; i < 2; i++) {
14+            for (int j = 0; j < 2; j++) {
15+                c[i][j] = a[i][j] + b[i][j];
16+            }
17+        }
18+        System.out.println("Matrix Addition:");
19+        for (int i = 0; i < 2; i++) {
20+            for (int j = 0; j < 2; j++) {
21+                System.out.print(c[i][j] + " ");
22+            }
23+            System.out.println();
24+        }
25+    }
26+ }
```

Output

Matrix Addition:
6 8
10 12

=== Code Execution Successful ===